

INCHEON INTL, Korea, Republic of

WMO: 471131

Lat: 37.469N

Lon: 126.451E

Elev: 23

StdP: 14.68

Time Zone: 9.00 (E09)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 12.3	(c) 15.8	(d) -5.8	(e) 4.0	(f) 17.9	(g) -2.4	(h) 4.8	(i) 19.8	(j) 26.0	(k) 20.7	(l) 23.3	(m) 22.7	(n) 10.9	(o) 320	(p) 0.485

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 10.3	(c) 89.2	(d) 75.8	(e) 86.2	(f) 75.2	(g) 84.2	(h) 74.1	(i) 78.8	(j) 83.5	(k) 77.7	(l) 82.8	(m) 76.5	(n) 81.8	(o) 10.0	(p) 300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.2	141.8	81.2	75.6	134.2	80.3	75.0	131.4	80.0	42.4	84.0	40.9	81.7	39.8	82.4	82.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 23.9	(b) 21.0	(c) 18.5	(e) 9.1	(f) 92.7	(g) 4.9	(h) 1.9	(i) 5.5	(j) 94.1	(k) 2.7	(l) 95.1	(m) -0.1	(n) 96.2	(o) -3.7	(p) 97.5		
			DB	WB	DB	WB	DB	WB	DB	WB	DB	WB	DB	WB		
			9.1	92.7	4.9	1.9	5.5	94.1	2.7	95.1	-0.1	96.2	-3.7	97.5		
			6.7	79.7	4.4	1.7	3.5	80.9	0.9	81.8	-1.5	82.8	-4.7	84.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	54.5	27.9	31.9	41.3	51.8	62.3	70.5	76.7	79.4	71.7	60.4	46.6	32.3		
	DBStd	18.48	6.82	6.75	5.76	5.26	4.39	3.46	3.46	3.95	3.92	5.75	7.79	7.92		
	HDD50	2219	686	508	277	40	0	0	0	0	0	5	155	548		
	HDD65	5054	1151	928	736	397	108	3	0	0	4	161	553	1013		
	CDD50	3863	0	0	6	94	380	616	826	911	650	328	52	0		
	CDD65	1222	0	0	0	1	23	169	361	446	203	19	0	0		
	CDH74	7890	0	0	0	7	92	626	2285	4019	828	33	0	0		
	CDH80	1837	0	0	0	1	4	65	495	1184	87	1	0	0		
(14) Wind	WSAvg	8.5	8.4	8.8	9.7	9.0	8.2	7.3	8.1	8.2	7.7	8.2	9.2	9.3		
(15) Precipitation	PrecAvg	53.5	0.7	1.1	1.5	3.0	4.1	5.5	15.7	11.5	5.9	2.0	1.8	0.9		
	PrecMax	87.0	2.0	2.4	4.2	6.5	10.4	19.8	37.7	30.9	22.0	8.4	4.3	2.5		
	PrecMin	29.7	0.1	0.0	0.2	0.7	0.6	1.3	6.5	1.6	0.5	0.1	0.4	0.1		
	PrecStd	12.9	0.6	0.7	1.0	1.3	2.2	4.0	7.0	7.5	5.2	1.8	0.9	0.6		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.7	55.2	62.7	72.0	80.3	84.4	91.3	93.5	84.6	77.4	66.5	53.7		
		MCWB	41.4	47.3	51.5	56.8	61.7	67.0	74.0	75.1	71.7	64.2	58.8	47.0		
	2%	DB	42.9	48.4	57.2	67.6	76.8	82.0	86.4	90.0	82.4	73.7	64.2	49.8		
		MCWB	38.4	42.3	49.1	54.4	61.4	67.2	75.0	75.9	69.9	62.9	57.4	44.7		
	5%	DB	40.7	44.7	53.5	64.3	73.5	80.2	84.4	88.0	80.5	71.5	60.9	46.6		
		MCWB	35.9	39.1	46.2	52.2	59.6	66.9	74.0	76.5	68.9	61.7	54.6	42.0		
	10%	DB	37.7	41.4	50.1	60.9	71.3	77.4	82.5	86.0	78.7	69.5	58.7	44.3		
		MCWB	33.4	36.5	43.3	50.8	58.6	66.3	73.4	75.5	67.6	60.6	52.3	39.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	42.2	49.7	54.5	60.3	66.9	74.6	78.8	80.7	76.7	68.8	61.1	49.3		
		MCDB	45.2	53.5	59.9	68.4	73.2	79.3	83.3	85.9	81.1	72.8	65.3	52.0		
	2%	WB	39.1	42.4	50.8	56.9	64.1	72.1	77.4	79.5	74.0	66.6	58.6	45.0		
		MCDB	42.2	46.0	55.3	64.0	70.5	76.6	82.8	84.4	77.6	71.0	62.4	48.7		
	5%	WB	36.4	39.7	47.2	54.6	62.4	70.2	76.3	78.6	72.0	64.5	55.3	42.3		
		MCDB	39.5	43.5	51.6	60.6	69.6	75.4	81.6	83.4	77.0	70.1	59.8	45.8		
	10%	WB	34.2	37.4	44.2	52.2	60.9	68.8	75.4	77.7	70.5	62.0	52.7	39.7		
		MCDB	37.4	41.2	48.3	58.6	67.9	74.2	80.2	82.7	76.1	67.4	57.9	43.3		
(35) Mean Daily Temperature Range	MDBR		12.3	13.1	13.8	14.7	15.2	13.1	10.0	10.3	12.0	13.4	12.5	12.3		
	5% DB	MCDBR	13.8	17.1	17.5	19.5	18.8	15.8	12.7	12.4	13.7	14.7	13.2	14.5		
		MCWBR	11.1	11.8	10.5	9.3	8.1	6.2	5.3	5.2	6.4	8.2	10.1	12.2		
	5% WB	MCDBR	13.2	15.6	15.5	16.1	15.4	11.8	10.5	10.1	11.1	11.9	12.6	13.6		
		MCWBR	11.6	12.1	11.1	9.9	8.4	6.3	5.4	5.2	6.6	8.6	11.1	12.6		
(40) Clear-Sky Solar Irradiance	taub		0.439	0.492	0.570	0.615	0.630	0.705	0.671	0.584	0.516	0.508	0.489	0.440		
	taud		1.952	1.857	1.722	1.676	1.698	1.620	1.736	1.942	2.047	1.934	1.926	1.967		
	Ebn at Noon		220	225	220	220	220	204	210	226	234	218	202	209		
	Edh at Noon		44	55	69	76	76	82	72	58	49	50	45	41		
(44) All-Sky Solar Radiation	RadAvg		798	1044	1385	1608	1802	1684	1335	1430	1347	1156	777	689		
	RadStd		50	82	131	103	174	193	208	200	138	80	75	44		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (39 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+164	+95

Nomenclature: See separate page